Experiment-3:

To aim for a more authentication and authorization setup using Angular, especially for a login form

Main.Ts

```
import { bootstrapApplication } from '@angular/platform-browser';
import { appConfig } from './app/app.config';
import { AppComponent } from './app/app.component';
bootstrapApplication(AppComponent, appConfig)
  .catch((err: unknown) => console.error(err));
```

app.component.html

```
<h1>Authentication Application</h1>
<router-outlet></router-outlet>
```

app.component.ts

```
import { Component } from '@angular/core';
import { RouterOutlet } from '@angular/router';
@Component({
  selector: 'app-root',
  standalone: true,
  imports: [RouterOutlet],
  templateUrl: './app.component.html'
})
export class AppComponent {
}
```

App.config.ts

```
import { ApplicationConfig } from '@angular/core';
import { provideRouter } from '@angular/router';
import { routes } from './app.routes';
export const appConfig: ApplicationConfig = {
  providers: [
    provideRouter(routes)
  ]
};
```

app.routes.ts

```
import { Routes } from '@angular/router';
import { LoginComponent } from './login/login.component';
import { HomeComponent } from './home/home.component';
import { authGuard } from './auth.guard';
export const routes: Routes = [
  {
    path: '',
    redirectTo: 'login',
    pathMatch: 'full'
  },
  {
    path: 'login',
    component: LoginComponent
  },
  {
    path: 'home',
    component: HomeComponent,
    canActivate: [authGuard]
  },
  {
```

```
    path: '**',
    redirectTo: 'login'
  }
];
```

auth.guard.ts

```
import { inject } from '@angular/core';
import { CanActivateFn, Router } from '@angular/router';
import { AuthService } from './auth.service';
export const authGuard: CanActivateFn = () => {
  const authService = inject(AuthService);
  const router = inject(Router);
  if (authService.isLoggedIn()) {
    return true;
  }
  router.navigate(['/login']);
  return false;
};
```

auth.service.ts

```
import { Injectable } from '@angular/core';
@Injectable({
  providedIn: 'root'
})
export class AuthService {
  login(username: string, password: string): boolean {
    if (username === 'admin' && password === '1234') {
      localStorage.setItem('isLoggedIn', 'true');
      return true;
    }
    return false;
  }
  logout(): void {
    localStorage.removeItem('isLoggedIn');
  }
  isLoggedIn(): boolean {
    return localStorage.getItem('isLoggedIn') === 'true';
  }
}
```

login.component.ts

```
import { Component, inject } from '@angular/core';
import { FormsModule } from '@angular/forms';
import { CommonModule } from '@angular/common';
import { Router } from '@angular/router';
import { AuthService } from '../auth.service';
@Component({
  selector: 'app-login',
  standalone: true,
  imports: [FormsModule, CommonModule],
  templateUrl: './login.component.html',
```

```
  styleUrl: './login.component.css'
})
export class LoginComponent {
  username = '';
  password = '';
  errorMessage = '';
  private authService = inject(AuthService);
  private router = inject(Router);
  login() {
    const success = this.authService.login(
      this.username,
      this.password
    );
    if (success) {
      this.errorMessage = '';
      // Go to Home page
      this.router.navigate(['/home']);
    } else {
      this.errorMessage = 'Invalid username or password';
    }
  }
}
```

login.component.html

```
<div class="login-container">
  <div class="login-box">
    <h2>Login</h2>
    <form (ngSubmit)="login()">
      <div class="form-group">
        <label>Username</label>
        <input
          type="text"
          [(ngModel)]="username"
          name="username"
          placeholder="Enter username"
          required>
      </div>
      <div class="form-group">
        <label>Password</label>
        <input type="password" [(ngModel)]="password" name="password" placeholder="Enter
password" required>
      </div>
      <button type="submit" class="login-btn"> Login </button>
    </form>
    @if (errorMessage) {
      <div class="error-message">
        {{ errorMessage }}
      </div>
    }
  </div>
</div>
```

login.component.css

```
/* Login Page */
.login-container {
  min-height: 100vh;
  display: flex;
  justify-content: center;
  align-items: center;
  background: linear-gradient(135deg, #667eea, #764ba2);
}
/* Login Card */
.login-box {
  width: 350px;
  padding: 35px;
  background-color: white;
  border-radius: 12px;
  box-shadow: 0 10px 30px rgba(0, 0, 0, 0.25);
}
/* Heading */
.login-box h2 {
  text-align: center;
  margin-bottom: 25px;
  color: #333;
  font-size: 28px;
}
/* Form Group */
.form-group {
  margin-bottom: 20px;
}
/* Labels */
.form-group label {
  display: block;
  margin-bottom: 7px;
```

```
  color: #444;
  font-weight: bold;
}
/* Input Fields */
.form-group input {
  width: 100%;
  padding: 12px;
  box-sizing: border-box;
  border: 1px solid #ccc;
  border-radius: 6px;
  font-size: 15px;
}
/* Input Focus */
.form-group input:focus {
  outline: none;
  border-color: #667eea;
  box-shadow: 0 0 5px rgba(102, 126, 234, 0.4);
}
/* Login Button */
.login-btn {
  width: 100%;
  padding: 12px;
  border: none;
  border-radius: 6px;
  background-color: #667eea;
  color: white;
  font-size: 16px;
  font-weight: bold;
  cursor: pointer;
  transition: 0.3s;
}
```

```
.login-btn:hover {
  background-color: #5568d9;
}
/* Error Message */
.error-message {
  margin-top: 15px;
  padding: 10px;
  background-color: #ffe5e5;
  color: #d00000;
  border-radius: 5px;
  text-align: center;
}
/* Responsive */
@media (max-width: 480px) {
  .login-box {
    width: 85%;
    padding: 25px;
  }
}
```

home.component.ts

```
import { Component, inject } from '@angular/core';
import { Router } from '@angular/router';
import { AuthService } from '../auth.service';
@Component({
  selector: 'app-home',
  standalone: true,
  imports: [],
  templateUrl: './home.component.html',
  styleUrl: 'home.component.css'
})
export class HomeComponent {
```

```
  private authService = inject(AuthService);
  private router = inject(Router);
  logout() {
    this.authService.logout();
    this.router.navigate(['/login']);
  }
}
```

home.component.html

```
<div class="home-container">
  <nav class="navbar">
    <h2>My Application</h2>
    <button class="logout-btn" (click)="logout()"> Logout  </button>
  </nav>
  <main class="home-content">
    <div class="welcome-box">
      <h1>Welcome!</h1>
      <p>You have successfully logged in.</p>
      <p>Welcome to the Home Page.</p>
    </div>
  </main>
</div>
```

home.component.css

```
/* Home Page */
.home-container {
  min-height: 100vh;
  background-color: #f4f6f9;
}
```

```
/* Navigation Bar */
.navbar {
  height: 65px;
  display: flex;
  justify-content: space-between;
  align-items: center;
  padding: 0 30px;
  background: linear-gradient(135deg, #667eea, #764ba2);
  color: white;
}
/* Website Title */
.navbar h2 {
  margin: 0;
  font-size: 24px;
}
/* Logout Button */
.logout-btn {
  padding: 10px 20px;
  border: none;
  border-radius: 6px;
  background-color: white;
  color: #667eea;
  font-weight: bold;
  cursor: pointer;
  transition: 0.3s;
}

.logout-btn:hover {
  background-color: #eeeeee;
}

/* Main Content */
```

```
.home-content {
  display: flex;
  justify-content: center;
  align-items: center;
  min-height: calc(100vh - 65px);
}

/* Welcome Card */
.welcome-box {
  width: 500px;
  padding: 40px;
  background-color: white;
  text-align: center;
  border-radius: 12px;
  box-shadow: 0 8px 25px rgba(0, 0, 0, 0.15);
}

/* Welcome Heading */
.welcome-box h1 {
  color: #333;
  margin-bottom: 15px;
}

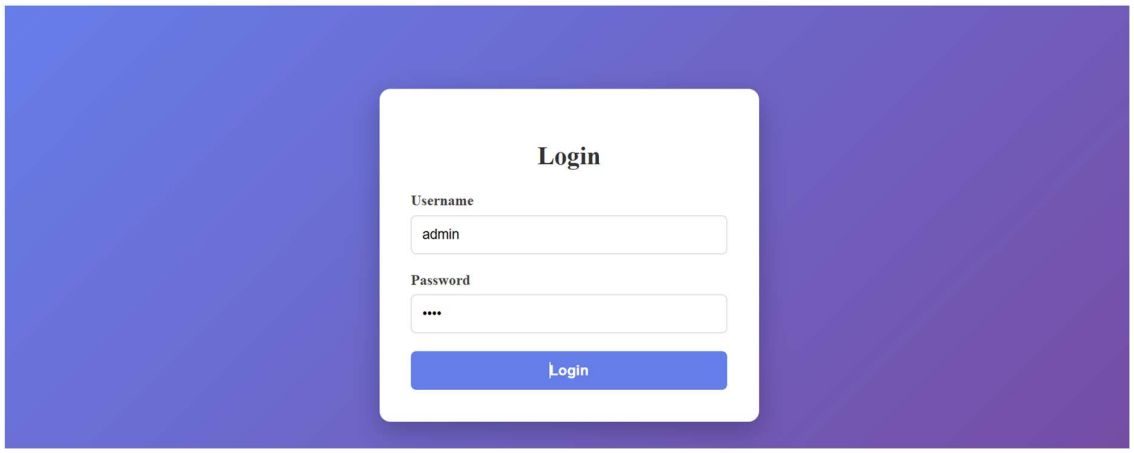
/* Welcome Text */
.welcome-box p {
  color: #666;
  font-size: 17px;
  line-height: 1.6;
}

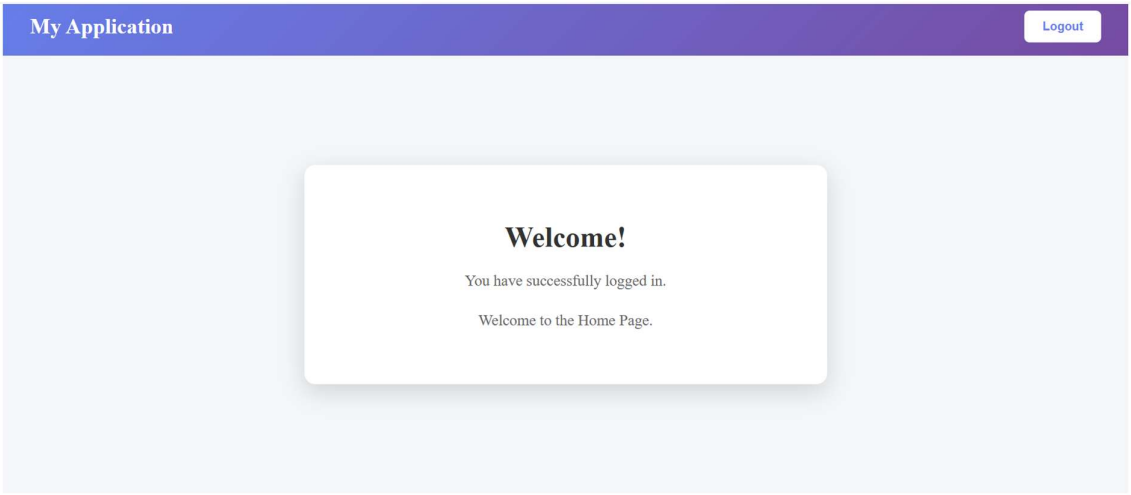
/* User Name */
.username {
```

```
  color: #667eea;
  font-weight: bold;
}

/* Responsive */
@media (max-width: 600px) {
  .navbar {
    padding: 0 15px;
  }
  .navbar h2 {
    font-size: 19px;
  }
  .welcome-box {
    width: 80%;
    padding: 25px;
  }
}
```

Authentication Application




My Application
Logout
Welcome!
You have successfully logged in.
Welcome to the Home Page.